a

Model variant	Global best (%)	Interface best (%)	Δ (iface–overall, %)	Iface Δ vs full (%, P)
Full model (ens2)	74.58	85.60	+10.05	—
– ensemble (single)	71.42	81.35	+9.08	-4.25 (P < 0.001)
– SS conditioning	70.59	81.12	+9.80	-4.48 (P < 0.001)
– data augmentation	70.66	79.29	+7.93	-6.31 (P < 0.001)
– ligand context	70.78	80.92	+8.71	-4.68 (P < 0.001)

300-sample pools (3 T × 100) · final noiface recipe · global = test 477 chains
interface = 115 ligand-contact chains (<5 Å; 100 with ≥3 interface residues);
rows 2–5 are single models — within that family, component removals cost
0.6–0.8% globally (interface effects n.s. at this pool size)









